

For example, let

$$A = \begin{bmatrix} 3 & 1 \\ -5 & x \end{bmatrix}, \text{ then } -A \text{ is given by}$$

$$-A = (-1)A = (-1) \begin{bmatrix} 3 & 1 \\ -5 & x \end{bmatrix} = \begin{bmatrix} -3 & -1 \\ 5 & -x \end{bmatrix}$$

Difference of matrices If $A = [a_{ij}]$, $B = [b_{ij}]$ are two matrices of the same order, say $m \times n$, then difference $A - B$ is defined as a matrix $D = [d_{ij}]$, where $d_{ij} = a_{ij} - b_{ij}$, for all value of i and j . In other words, $D = A - B = A + (-1)B$, that is sum of the matrix A and the matrix $-B$.

Example 7 If $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & -1 & 3 \\ -1 & 0 & 2 \end{bmatrix}$, then find $2A - B$.

Solution We have

$$\begin{aligned} 2A - B &= 2 \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix} - \begin{bmatrix} 3 & -1 & 3 \\ -1 & 0 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 2 & 4 & 6 \\ 4 & 6 & 2 \end{bmatrix} + \begin{bmatrix} -3 & 1 & -3 \\ 1 & 0 & -2 \end{bmatrix} \\ &= \begin{bmatrix} 2-3 & 4+1 & 6-3 \\ 4+1 & 6+0 & 2-2 \end{bmatrix} = \begin{bmatrix} -1 & 5 & 3 \\ 5 & 6 & 0 \end{bmatrix} \end{aligned}$$

3.4.3 Properties of matrix addition

The addition of matrices satisfy the following properties:

- (i) **Commutative Law** If $A = [a_{ij}]$, $B = [b_{ij}]$ are matrices of the same order, say $m \times n$, then $A + B = B + A$.

$$\begin{aligned} \text{Now } A + B &= [a_{ij}] + [b_{ij}] = [a_{ij} + b_{ij}] \\ &= [b_{ij} + a_{ij}] \text{ (addition of numbers is commutative)} \\ &= ([b_{ij}] + [a_{ij}]) = B + A \end{aligned}$$

- (ii) **Associative Law** For any three matrices $A = [a_{ij}]$, $B = [b_{ij}]$, $C = [c_{ij}]$ of the same order, say $m \times n$, $(A + B) + C = A + (B + C)$.

$$\begin{aligned} \text{Now } (A + B) + C &= ([a_{ij}] + [b_{ij}]) + [c_{ij}] \\ &= [a_{ij} + b_{ij}] + [c_{ij}] = [(a_{ij} + b_{ij}) + c_{ij}] \\ &= [a_{ij} + (b_{ij} + c_{ij})] \quad (\text{Why?}) \\ &= [a_{ij}] + [(b_{ij} + c_{ij})] = [a_{ij}] + ([b_{ij}] + [c_{ij}]) = A + (B + C) \end{aligned}$$

- (iii) **Existence of additive identity** Let $A = [a_{ij}]$ be an $m \times n$ matrix and O be an $m \times n$ zero matrix, then $A + O = O + A = A$. In other words, O is the additive identity for matrix addition.
- (iv) **The existence of additive inverse** Let $A = [a_{ij}]_{m \times n}$ be any matrix, then we have another matrix as $-A = [-a_{ij}]_{m \times n}$ such that $A + (-A) = (-A) + A = O$. So $-A$ is the additive inverse of A or negative of A .

3.4.4 Properties of scalar multiplication of a matrix

If $A = [a_{ij}]$ and $B = [b_{ij}]$ be two matrices of the same order, say $m \times n$, and k and l are scalars, then

$$(i) \quad k(A + B) = kA + kB, \quad (ii) \quad (k + l)A = kA + lA$$

$$(ii) \quad k(A + B) = k([a_{ij}] + [b_{ij}])$$

$$= k[a_{ij} + b_{ij}] = [k(a_{ij} + b_{ij})] = [(ka_{ij}) + (kb_{ij})]$$

$$= [ka_{ij}] + [kb_{ij}] = k[a_{ij}] + k[b_{ij}] = kA + kB$$

$$(iii) \quad (k + l)A = (k + l)[a_{ij}]$$

$$= [(k + l)a_{ij}] = [ka_{ij}] + [la_{ij}] = k[a_{ij}] + l[a_{ij}] = kA + lA$$

Example 8 If $A = \begin{bmatrix} 8 & 0 \\ 4 & -2 \\ 3 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -2 \\ 4 & 2 \\ -5 & 1 \end{bmatrix}$, then find the matrix X , such that

$$2A + 3X = 5B.$$

Solution We have $2A + 3X = 5B$

$$\text{or} \quad 2A + 3X - 2A = 5B - 2A$$

$$\text{or} \quad 2A - 2A + 3X = 5B - 2A \quad (\text{Matrix addition is commutative})$$

$$\text{or} \quad O + 3X = 5B - 2A \quad (-2A \text{ is the additive inverse of } 2A)$$

$$\text{or} \quad 3X = 5B - 2A \quad (O \text{ is the additive identity})$$

$$\text{or} \quad X = \frac{1}{3} (5B - 2A)$$

$$\text{or} \quad X = \frac{1}{3} \left(5 \begin{bmatrix} 2 & -2 \\ 4 & 2 \\ -5 & 1 \end{bmatrix} - 2 \begin{bmatrix} 8 & 0 \\ 4 & -2 \\ 3 & 6 \end{bmatrix} \right) = \frac{1}{3} \left(\begin{bmatrix} 10 & -10 \\ 20 & 10 \\ -25 & 5 \end{bmatrix} + \begin{bmatrix} -16 & 0 \\ -8 & 4 \\ -6 & -12 \end{bmatrix} \right)$$